

Data Structure and Algorithms

BCA — Semester 3 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. How stack is used in recursion? Explain different stack operations. Explain algorithm to convert an infix expression to postfix using stack.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Explain complete binary tree with example. Starting with an empty binary search tree, show the effect of successively adding the following elements: 47, 50, 25, 27, 17, 61, 5, and 26. Also, traverse the resulting tree in pre-order, in-order, and post-order.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain quick sort algorithm. Use this algorithm to sort the numbers 35, 82, 18, 54, 13, 31, 20, 69, and 19.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Define ADT. Explain array as an ADT.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is time complexity? Explain big oh notation with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain priority queue with example. What is circular queue?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What are the benefits of using recursion? Write a recursive function to find nth Fibonacci number.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Explain singly linked list with example. Compare singly linked list with doubly linked list.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain different applications of binary tree.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain collision and collision resolution in hashing. What is double hashing?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Explain adjacency matrix representation of graphs with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on:
a. Linear Search
b. Minimum Spanning Tree

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.